

Page speed – the most important

It is recommended that page speed should be one of the first metrics that should be taken care of while building and maintaining your website, and it should be continuously monitored. Do it right at the onset and the website will definitely work wonders. This does not imply that you cannot optimise the page speeds for existing or maintenance websites.

There are a number of audits that Google runs, which rank your website for page speed based on the following:

1. Performance
2. PWA (progressive Web app)
3. Best practices
4. Accessibility
5. SEO

It also considers how your website loads on mobiles and averages your score out, so take care of smaller devices too. We have already covered accessibility and SEO.

Performance

Ever since the recent Google algorithm speed update, the performance of your website plays a crucial role in website rankings. Many websites saw a decrease in their page ranking scores around late 2018 and have realised the importance of 'performance' in website ranking.

The key metrics for performance are:

1. First Contentful Paint or FCP (Is it happening?)
2. First Meaningful Paint or FMP (Is it useful?)
3. Speed index (How quickly is it loaded?)
4. First CPU idle time (Can I interact?)
5. Time to be interactive (Can I interact with all components on the page?)
6. Maximum potential first input delay (How long does the longest task run?)

FCP: This is the time taken by

the website to show the first piece of content on the page. Google says that if users do not see something coming up on a website within two seconds, they drop out of the website.

FMP: This is the time taken for a page's primary content to come up on the screen. This can include any contentful elements like text, image, canvas, etc.

Speed index: This measures how quickly the contents get displayed during page load.

First CPU idle time: This measures how long it takes for the page to become minimally interactive.

Time to be interactive: This is the time taken by your website to become fully interactive.

Maximum potential first input delay: This measures the time from when a user first interacts by clicking or tapping, to the time when the browser is actually able to respond.

You can check out your website scores at <https://developers.google.com/speed/pagespeedinsights>.

Now that we have seen the key metrics, we need to improve on them. Some quick steps to achieve this are:

- Reduce redirects
- Use a content distribution network
- Project profile
- Use hero images
- Choose analytics scripts wisely
- Minimise your CSS
- Remove render-blocking JavaScript
- Leverage browser caching
- Optimise images (use .png and .jpeg efficiently, sprite images)
- Lazy load below the fold
- Use .woff2 instead of .woff for fonts
- Make use of a bundler
- Avoid FOIT (flash of invisible text)
- Use ES6

Other tools for measuring and

improving performance are:

- *Lighthouse* helps you measure your website page's speed from the developer tab on Google Chrome.
- *WebpageTest.org* runs numerous tests at a time on your live website and gets your field data.
- *Bundlephobia.com* helps you analyse the cost of adding a particular bundle in terms of its size.
- *Lazysizes* helps you optimise the loading of content below the fold.
- *Webfont generator* converts your .woff font files to .woff2 files, which are much lighter to load.
- *Pagespeedinsights* gives your website scores and highlights problem areas to be resolved.
- *PerfumeJS*: You can integrate this library, and add checks in your code to measure how much time a piece of code takes to execute.

Progressive Web apps (PWAs)

This is a website that looks and behaves like a mobile app. PWAs are built to take advantage of native mobile device features, without requiring the end user to visit an app store.

The core values of a good PWA are:

- Uses HTTPS
- Has a valid Web manifest
- Registers a service worker with a fetch event handler

Along with these, loading fast, working offline and being platform-integrated are equally expected.

Achieve this and you can score well in the PWA area too.

There are over 200 aspects to Google's page ranking criteria. We have covered just a few here, which we as developers can easily take care of to ensure a higher Google page ranking. Hope this information helps you in some way. **END** 

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